AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 7 · STUDENT LESSON

Testing pure components

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Least privilege and tool scopes. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of testing pure components. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of testing pure components.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of testing pure components

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Least privilege and tool scopes. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain testing pure components using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Testing pure components case. Separate observations, given facts and assumptions before choosing an explanation.

Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Testing pure components. Connect each labelled input, deciding step and result.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

Testing pure components — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13.	Define remaining(capacity, used)=capacity-used for integer inputs satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.	Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Testing pure components. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Supply zero instead of 7.

Predicted result

Prediction: Reject the **input** under the stated positive-integer contract.

Why the result follows

Reason: Zero has the right type but violates the value requirement.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Testing pure components to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Pure function	A computation with no externally observable side effects
Boundary case	An input at a contract edge
Expected outcome	The result or error specified before execution

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Testing pure components, predict a step, then compare with its explanation.

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13.

Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Worked explanation · Testing pure components

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Testing pure components

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def remaining(capacity, used):
    if not nonnegative_integer(capacity) or not nonnegative_integer(used) or used >
capacity:
        raise ValueError('capacity/used')
    return capacity - used

print(remaining(12, 5))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Testing pure components

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

The facts we will use

Supplied facts

Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

AI ACADEMY / AI-6/7

Testing pure components

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Call with 7

The prefix and supplied numeral produce Room 7.

2. Compare exact text

The expected text includes one space before 7.

3. Call with 12

The same fixed procedure produces Room 12.

CHANGE ONE THING

Supply zero instead of 7.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Call with 7	The prefix and supplied numeral produce Room 7.
2. Compare exact text	The expected text includes one space before 7.
3. Call with 12	The same fixed procedure produces Room 12.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

What this example does not establish

Keep the stated limit

These tests cover only stated cases; they do not test the full AI service or every invalid value.

Keep the stated limit

Read the labelled picture: Predict the formatter **output**s for 7 and 12, including spacing. Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

Keep the stated limit

Change one thing: Supply zero instead of 7. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Testing pure components claim and explain a repair using the supplied evidence.

Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Testing pure components with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Testing pure components”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Testing pure components. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Predict the formatter **output**s for 7 and 12, including spacing. Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Testing pure components. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def remaining(capacity, used):
    if not nonnegative_integer(capacity) or not nonnegative_integer(used) or used >
capacity:
        raise ValueError('capacity/used')
    return capacity - used
```

```
print(remaining(12, 5))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Testing pure components. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its **expected outcome** before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled testing pure components **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing pure components.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	

Project evidence	Record
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its **expected outcome** before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Testing pure components case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Testing pure components — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13.	Define remaining(capacity, used)=capacity-used for integer inputs satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.	Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Testing pure components, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

By the end, I can	How I will show it
Explain the mechanism and terminology of testing pure components	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Testing pure components without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of testing pure components

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

> block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The same fixed procedure produces Room 12.

1. Call with 12

2. Compare exact text

3. Call with 7

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Call with 12 The same fixed procedure produces Room 12. A pure component depends on its accepted **input**, not a clock, network or model.

Isolating it makes failures easier to locate.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply zero instead of 7. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Reject the **input** under the stated positive-integer contract.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject the **input** under the stated positive-integer contract. Zero has the right type but violates the value requirement.

Repair a missing building block

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

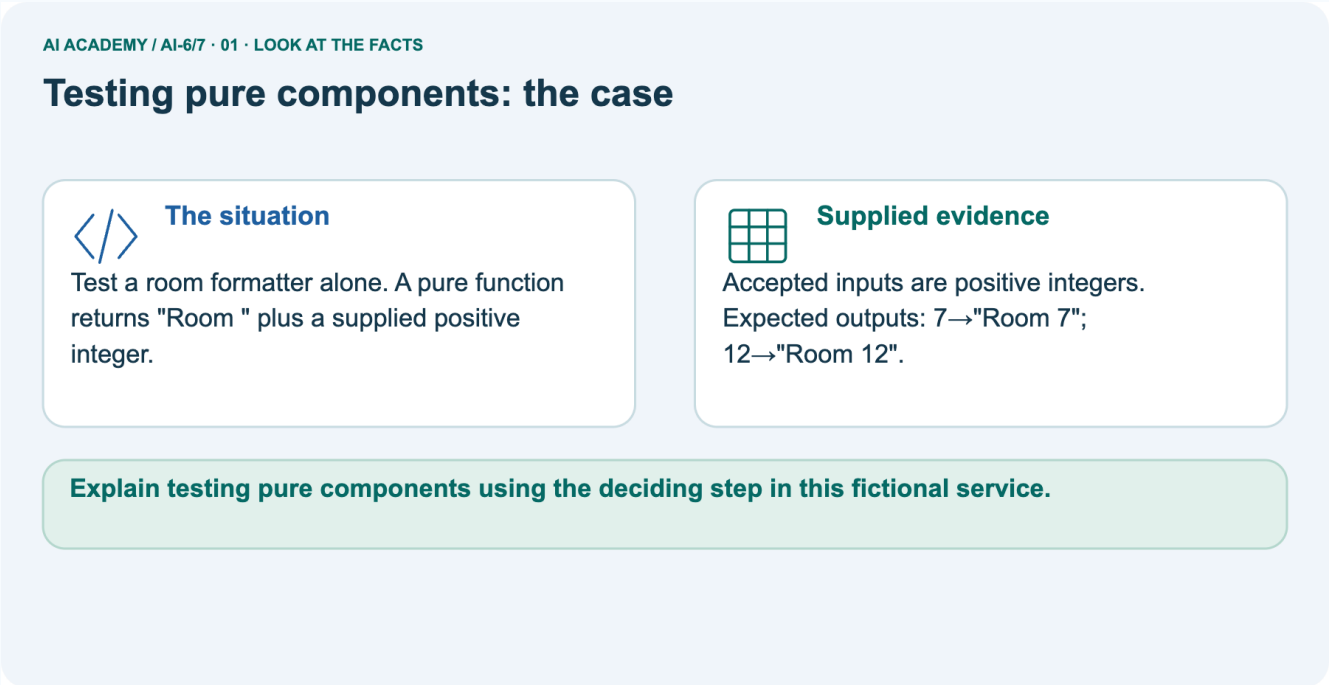
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Testing pure components graphic using its labels, deciding evidence and relevant limitation.

01. Testing pure components: the case

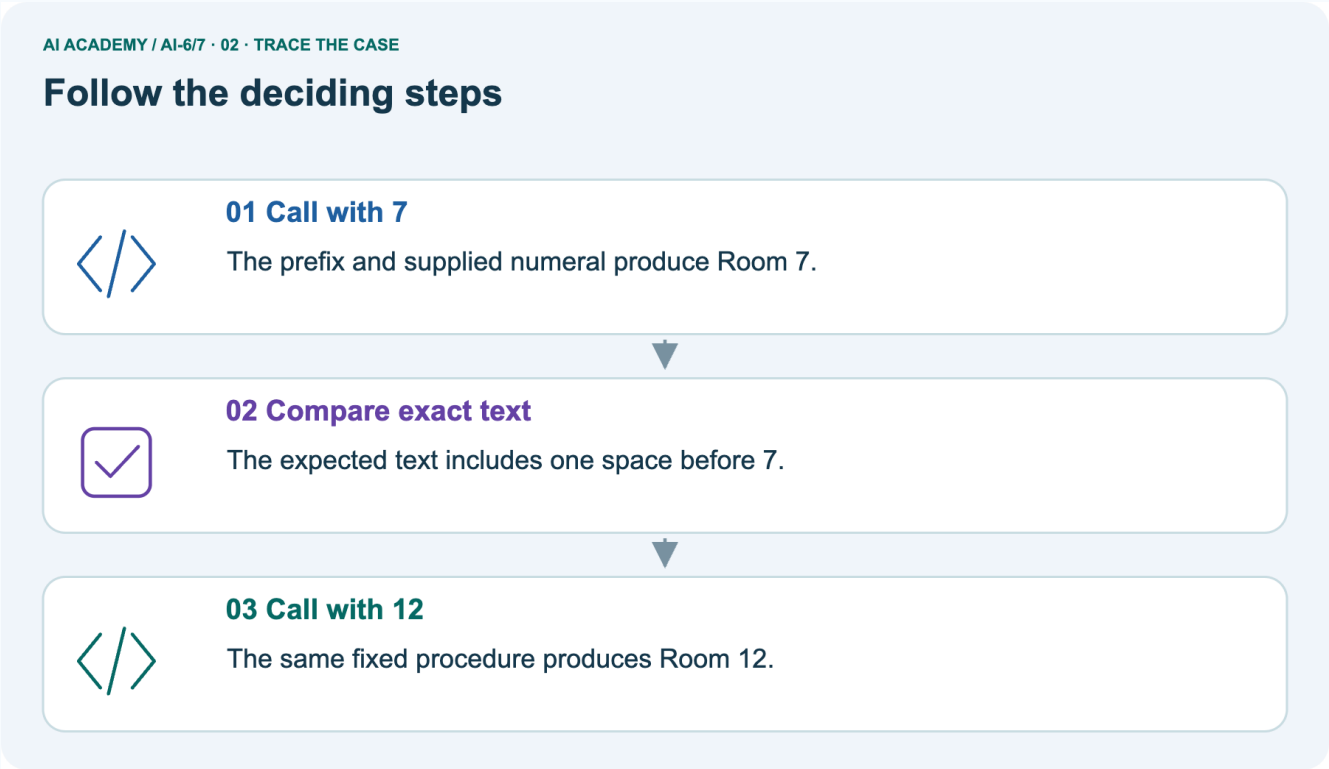


Follow the labelled picture with the matching explanation. Open the vector diagram

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer. Supplied facts: Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

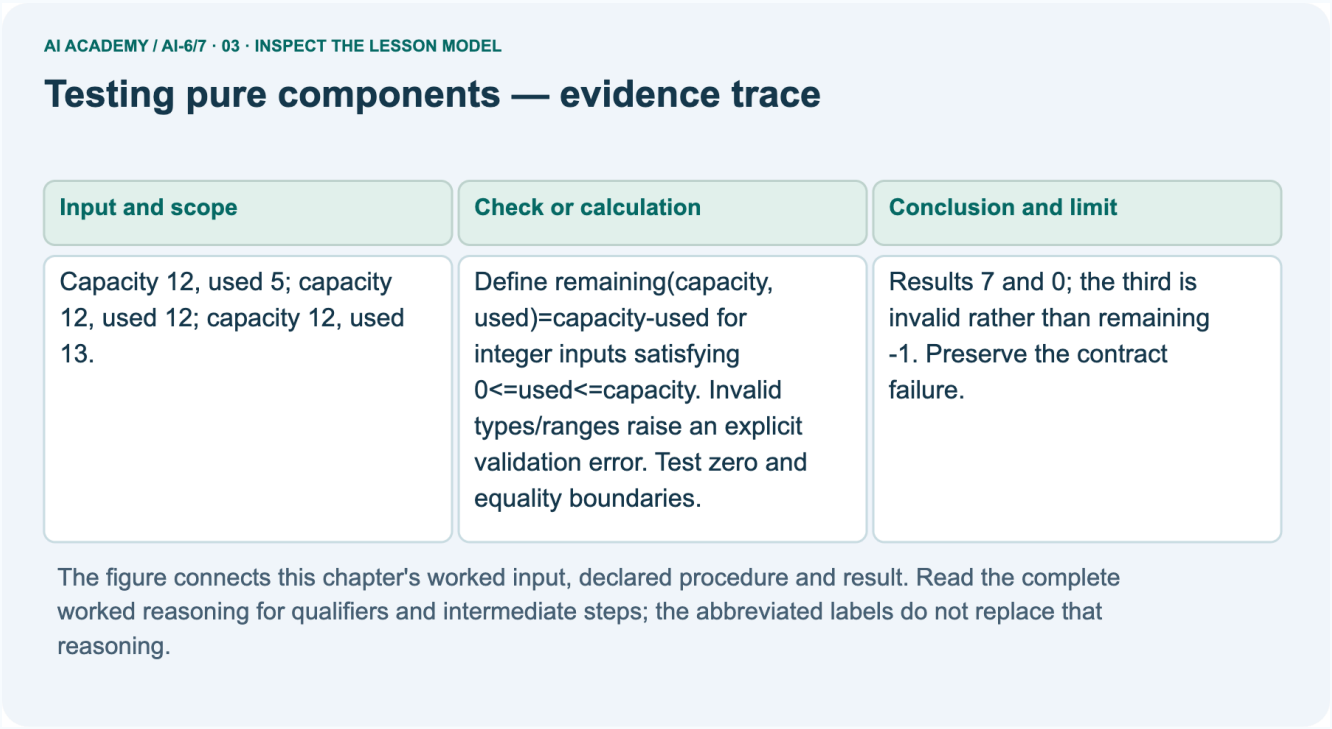


Follow the labelled picture with the matching explanation. Open the vector diagram

Call with 7: The prefix and supplied numeral produce Room 7. → Compare exact text: The expected text includes one space before 7. → Call with 12: The same fixed procedure produces Room 12.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Testing pure components — evidence trace



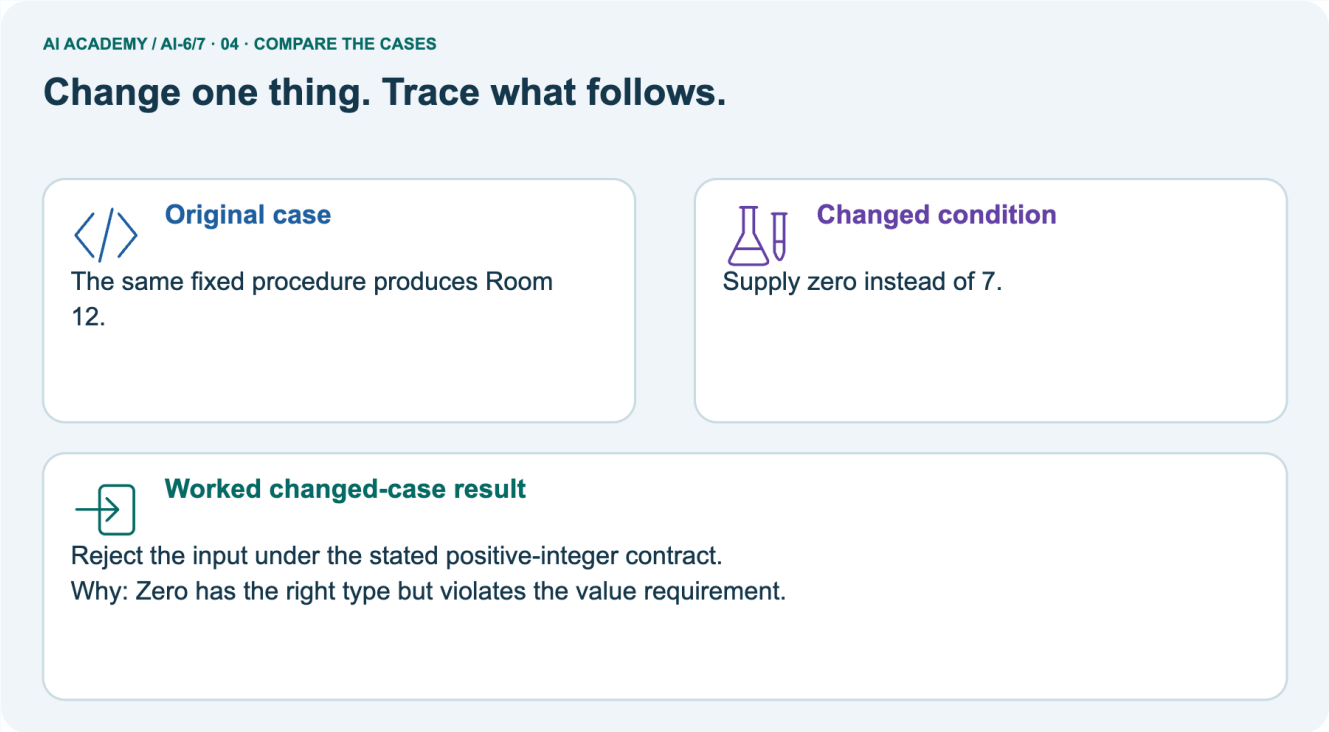
Follow the labelled picture with the matching explanation. Open the vector diagram

Testing pure components — **evidence** trace. Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. | Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity.

Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries. | Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



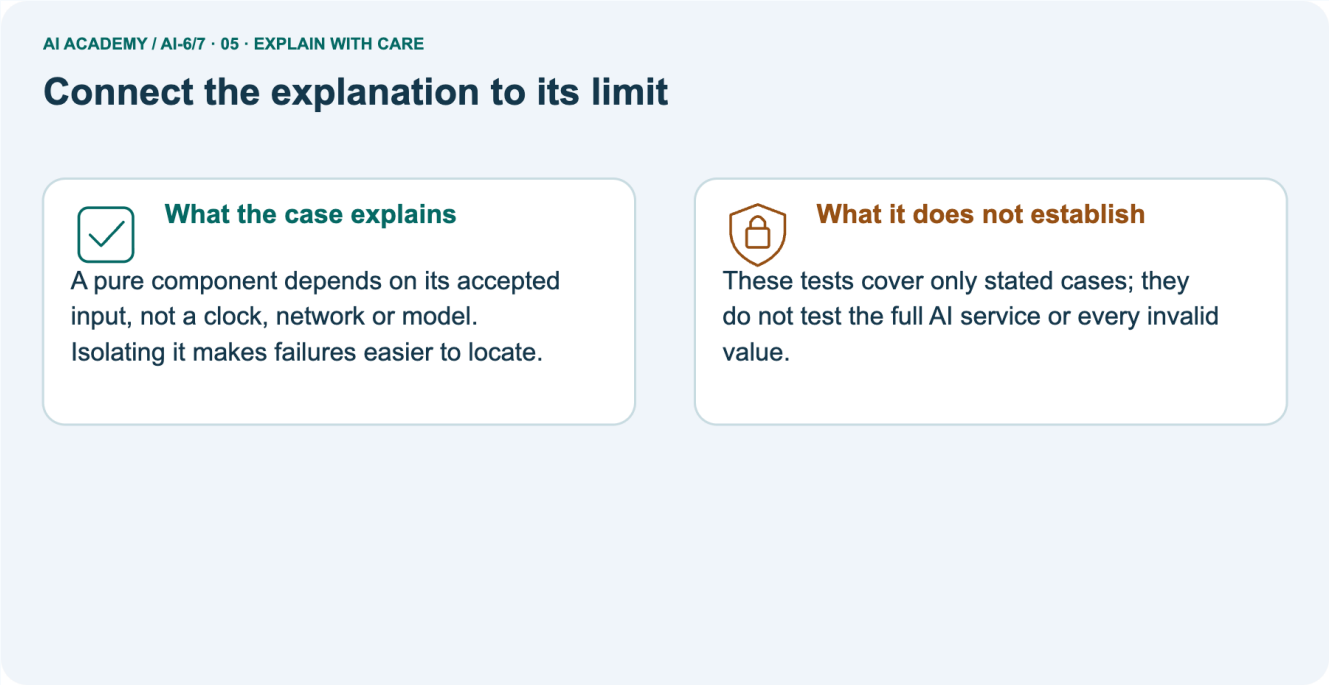
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The same fixed procedure produces Room 12. Changed condition: Supply zero instead of 7. Expected result: Reject the **input** under the stated positive-integer contract.

Reason: Zero has the right type but violates the value requirement.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



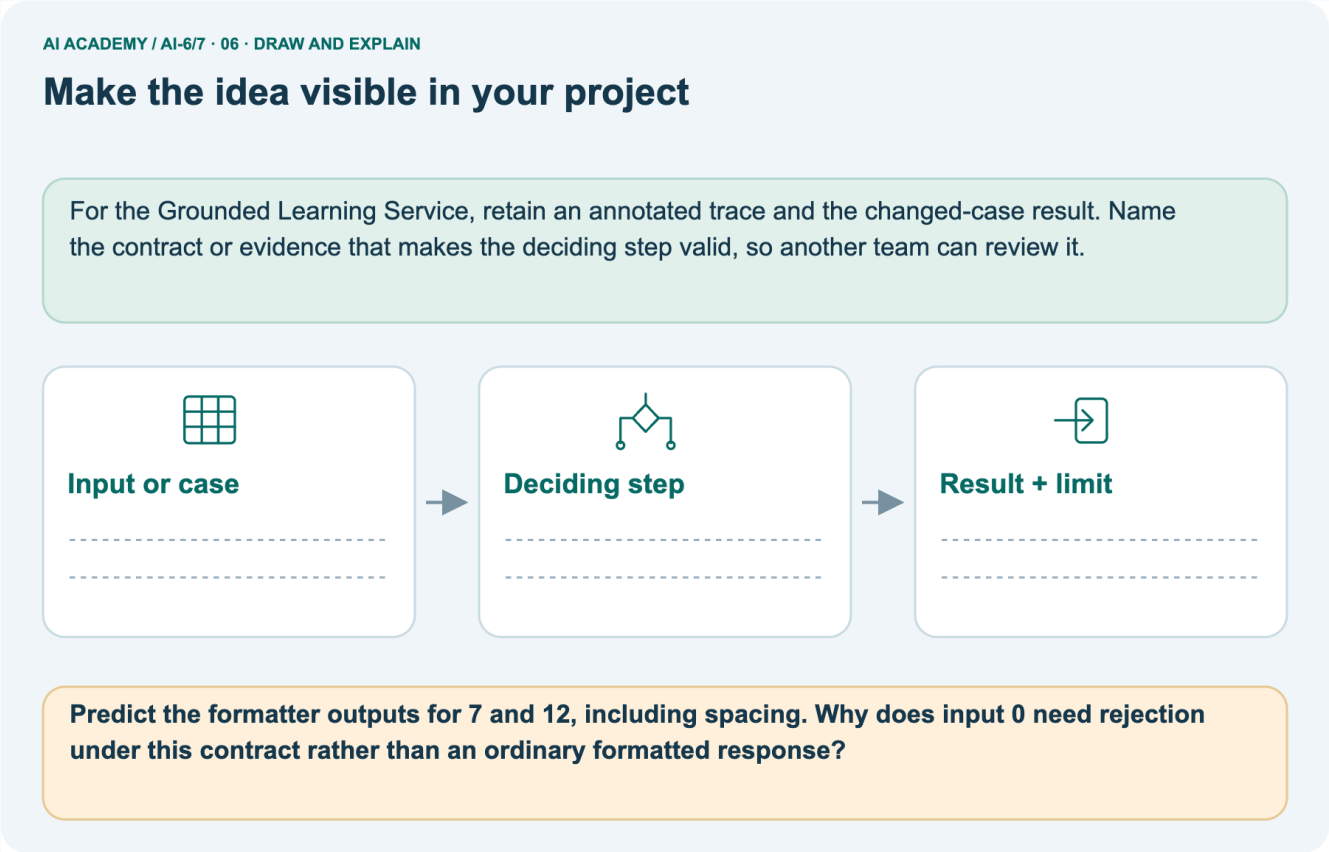
Follow the labelled picture with the matching explanation. Open the vector diagram

A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

Limit: These tests cover only stated cases; they do not test the full AI service or every invalid value.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.